

COMICCRAFT — AI COMIC STORY CREATOR

2. Requirement Analysis

FUNCTIONAL REQUIREMENTS

- FR-01 Collect story prompt, character name, setting, tone and art style.
- FR-02 Validate and process form / JSON input through FastAPI.
- FR-03 Generate a structured 5-panel comic outline.
- FR-04 Generate narration and character dialogue for each panel.
- FR-05 Generate comic-style illustrations from image prompts.
- FR-06 Build a panel-by-panel comic layout.
- FR-07 Display the generated comic in a web preview.
- FR-08 Export the complete comic as a downloadable PDF.
- FR-09 Provide API routes for comic generation and image

DATA / INPUT REQUIREMENTS

- Story Prompt — main idea for the comic.
- Character Name — hero / main character.
- Setting — forest, school, city, etc.
- Story Tone — dramatic, funny, poetic, light-hearted.
- Art Style — anime, pixel art, comic book, realistic.
- Generated data includes panel number, title, scene description, image prompt, narration and dialogue.
- Generated images are stored for panel layout and PDF export.

NON-FUNCTIONAL REQUIREMENTS

- Usability: simple, interactive HTML/jinja2 interface.
- Responsiveness: clean, readable layouts and organized result panels.
- Maintainability: modular generation, layout and export functions.
- Performance: efficient AI-assisted text and image generation.
- Reliability: backend error handling with FastAPI HTTPException / try-except.
- Security: Gemini and Hugging Face credentials stored through environment variables.
- Accessibility: designed for non-artists and casual users.

INTERFACE REQUIREMENTS

- index.html — collects comic creation inputs.
- comic_preview.html — displays generated panels sequentially.
- export_success.html — confirms successful PDF export.
- Preview should show panel title, image, scene description, captions and narration.
- Navigation should provide a direct flow from input → generation → preview → export.
- Jinja2 templates dynamically render AI-generated content.

TECHNOLOGY & EXTERNAL DEPENDENCIES

- FastAPI + Uvicorn
- HTML/CSS + Jinja2
- Google Gemini Flash / Pro
- Hugging Face Diffusers + Stable Diffusion
- PyTorch
- FPDF
- Pillow